

Product Requirements Document

Project: Green-IDS (Green Learning Intrusion Detection System)

1. Product Overview

The Green-IDS project aims to provide a sustainable, transparent, and high-performance alternative to traditional Deep Learning-based intrusion detection systems. By utilizing the Green Learning (GL) paradigm, this project addresses the limitations of current "black box" models, specifically their computational intensity and high energy consumption.

2. Objectives

- **Sustainability:** Minimize the carbon footprint through modular, non-iterative training.
- **Transparency:** Ensure clear, logical decision-making paths rather than black-box mappings.
- **Efficiency:** Maintain competitive accuracy with a significantly smaller model footprint.
- **Edge Readiness:** Target deployment in resource-constrained IoT and edge environments.

3. Architecture

The system follows the three-module Green Learning architecture:

- **Representation Learning:** Uses data-driven transforms (e.g., Saab/Saak) to concisely represent network traffic.
- **Supervised Feature Learning:** Implements Discriminant Feature Test (DFT) to isolate critical malicious indicators.
- **Supervised Decision Learning:** Uses Subspace Learning Machine (SLM) for lightweight classification.

4. Functional Requirements

Feature	Description	Priority
Data Preprocessing	Normalize UNSW-NB15 dataset	Must-have
Modular Pipeline	Execute the three-module GL flow	Must-have
Feature Selection	Implement DFT/RFT for dimensionality reduction	Must-have

Inference Engine	Lightweight classification module	Should-have
Sustainability Reporting	Track energy/parameter usage	Could-have

5. Technical Constraints

The solution must be built using Python, focusing on modularity. Transparency must be maintained by allowing inspection of feature selection at each stage.

6. Success Metrics

- **Accuracy:** F1-score parity with existing DL baselines.
- **Model Size:** Reduction of >90% in total parameters.
- **Latency:** Low-millisecond inference time.